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Connecting Wireless Laptops to a Wireless Router (WRT300N) in Cisco Packet Tracer

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Aim

To set up a wireless network by configuring a WRT300N wireless router with an SSID, WPA2-PSK security and a DHCP server, connecting wireless laptops to it, and verifying communication using ping.

Requirements

Cisco Packet Tracer; 1 × WRT300N Wireless Router; 2 or more Laptop-PT; WPC300N wireless adapter module for each laptop.

Network Details

Parameter	Value used
SSID	Home Router
Security / Encryption	WPA2-PSK / AES
PSK Pass Phrase	Password
Router (Gateway) IP	192.168.0.1
Subnet Mask	255.255.255.0
Internet Connection Type	Automatic Configuration – DHCP
DHCP Start IP / Max Users	192.168.0.2 / 50 (range 192.168.0.2 – 192.168.0.51)

Before You Start

On the Windows computer: File Explorer → Network → Properties → Change adapter settings → disable the active Ethernet and the active Wi-Fi adapters, so that they do not interfere with the Packet Tracer practical.

Procedure

1. Build the Network

Step 1: Open Cisco Packet Tracer.

Step 2: From Network Devices (bottom-left) → Wireless Devices, select the WRT300N Wireless Router0 and place it on the workspace.

Step 3: From End Devices (bottom-left), place the required number of Laptop-PT (here Laptop0 and Laptop1).

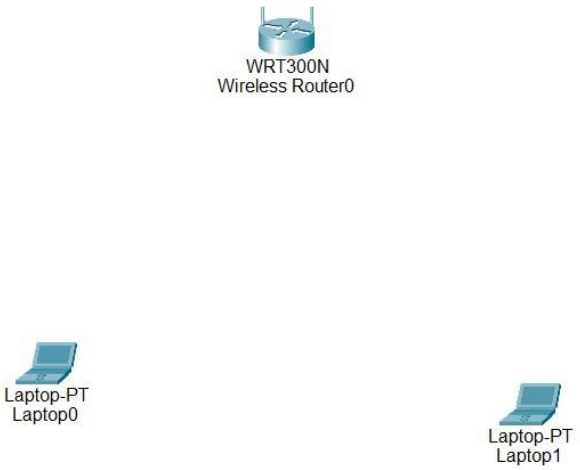


Fig 1: Wireless router and laptops placed in Packet Tracer

2. Configure Wireless Security on the Router

Step 4: Double-click the WRT300N router → Config tab → Wireless.

Step 5: Enter the SSID name (Home Router), select WPA2-PSK (or any other security), type the PSK pass phrase (Password), keep encryption as AES and close the window.

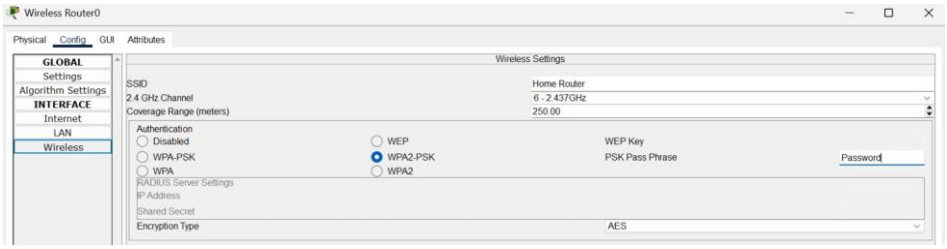


Fig 2: Wireless settings: SSID, WPA2-PSK and pass phrase

3. Configure DHCP and Gateway IP on the Router

Step 6: Open the router's GUI tab and make sure Internet Connection Type is Automatic Configuration – DHCP.

Step 7: Under Router IP set the Gateway IP address and subnet mask. The default is generally 192.168.0.1 / 255.255.255.0; it can be changed as required, e.g. 192.168.1.1 or 192.168.30.1.

Step 8: Under DHCP Server Settings keep DHCP Enabled and set the Start IP Address (e.g. 192.168.0.2 or 192.168.0.10, in the same range as the gateway, with the default mask 255.255.255.0).

Step 9: Set the Maximum number of Users (10, 20, 50 as required), scroll down and click Save Settings.

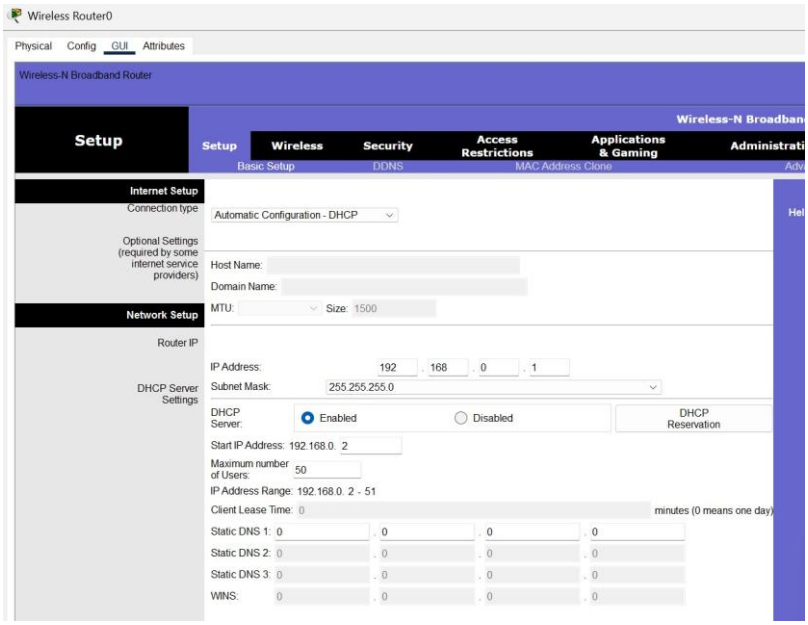


Fig 3: Router GUI: DHCP mode, gateway 192.168.0.1, DHCP server enabled

4. Install the Wireless Adapter in the Laptop

Step 10: Click Laptop0 → Physical tab. Switch OFF the laptop by clicking the power button.

Step 11: Drag the existing wired (LAN Ethernet) module out of its slot, and drag the WPC300N (wireless) module into the empty slot.

Step 12: Switch the laptop ON again by clicking the power button. The Linksys-WPC300N is now shown in the slot.



Fig 4: WPC300N wireless module installed in Laptop0

5. Connect the Laptop to the Wireless Router

Step 13: Go to Desktop → PC Wireless. Open the Connect tab (next to Link Information). It lists the available wireless network with its SSID (Home Router) and signal strength (100%).

Step 14: Select the network and click Connect (next to Refresh). When asked for the pre-shared key, enter the password given in the router (Password) and press Connect.

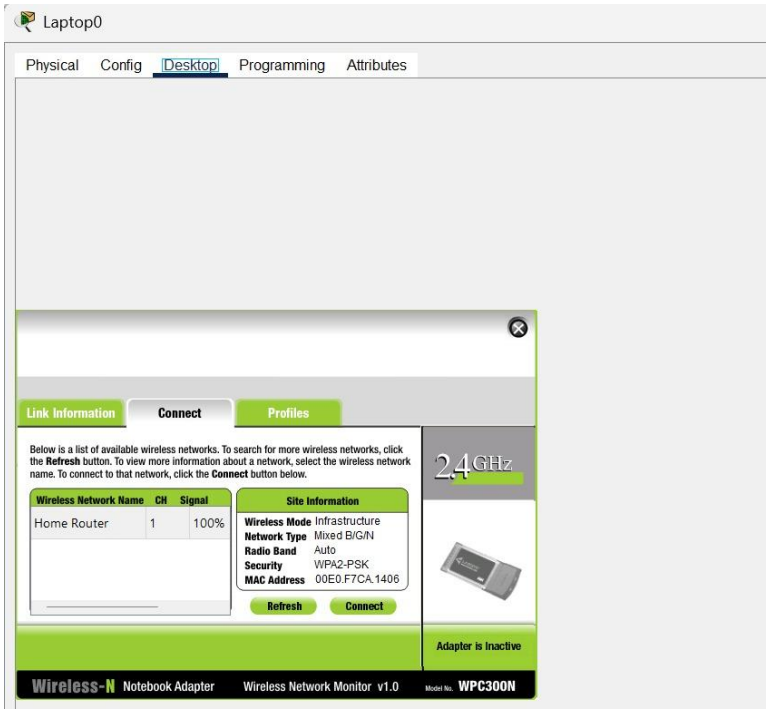


Fig 5: PC Wireless – Connect tab showing Home Router with 100% signal

The screen shows that Laptop0 is connected with the WRT300N router, and a wireless link appears between them on the workspace.

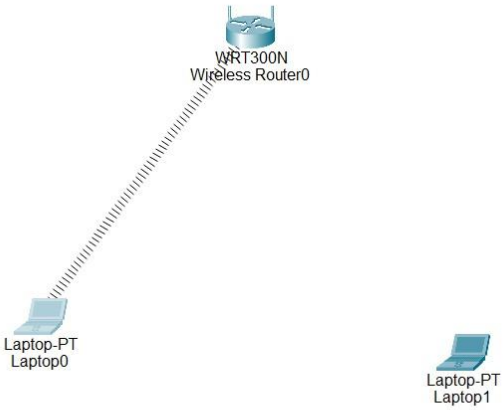


Fig 6: Laptop0 connected to the router (wireless link)

Step 15: Repeat Steps 4 and 5 for Laptop1, Laptop2 and so on, as many laptops as required.

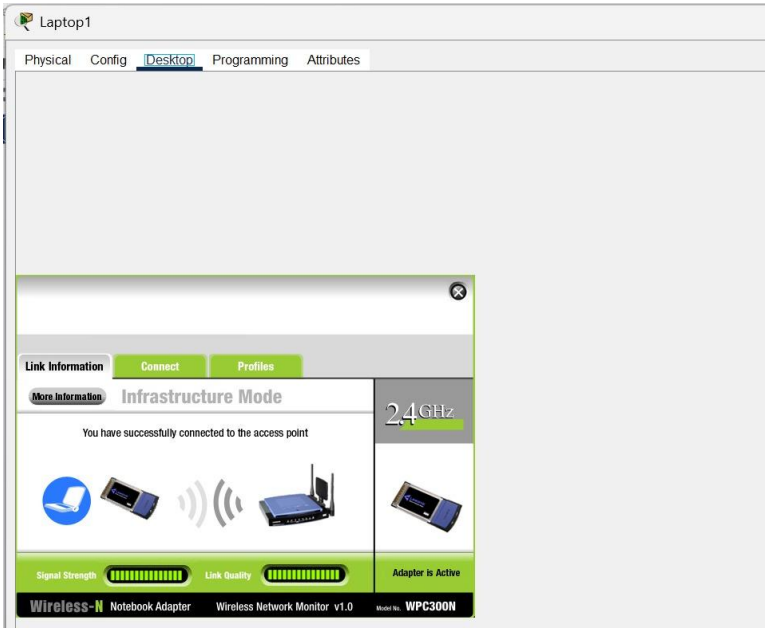


Fig 7: Laptop1 – Link Information: successfully connected to the access point

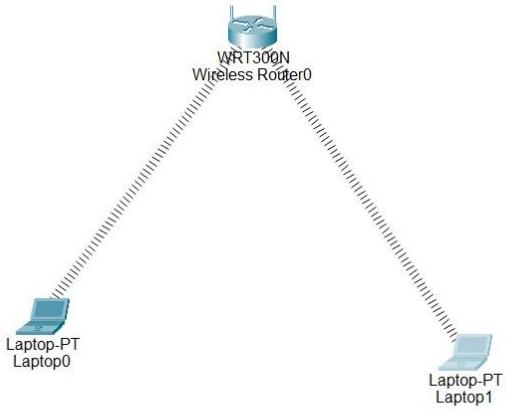


Fig 8: Both laptops connected to the wireless router

Verification

Step 16: Open Laptop0 → Desktop → Command Prompt. Ping the router's gateway IP (192.168.0.1 or whichever was set). Replies mean the laptop can reach the router.

```
C:\> ping 192.168.0.1
```

Step 17: Find the DHCP IP of the other laptop with ipconfig on Laptop1, then ping it from Laptop0 (for example ping 192.168.0.3 or 192.168.0.5). Ping the other laptops' DHCP IPs in the same way and check that each one replies.

```
C:\> ping 192.168.0.5
```

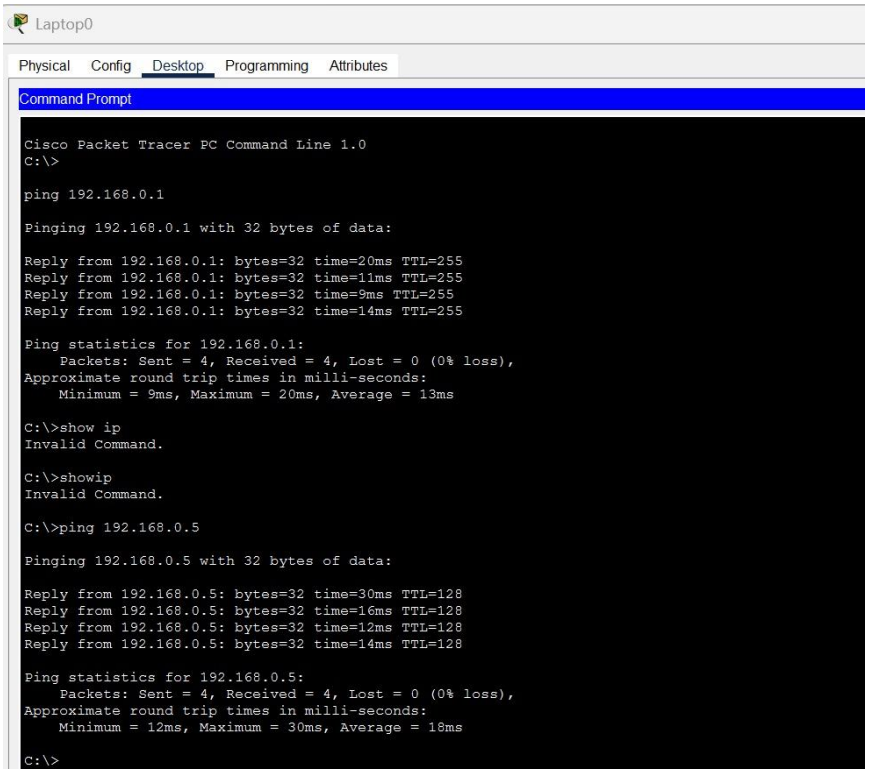


Fig 9: Ping from Laptop0 to the router (TTL=255) and to Laptop1 (TTL=128) – 0% loss

Note: “show ip” is not a valid command on the PC prompt (see Invalid Command in the screenshot); use ipconfig instead.

Result

✓ EXPERIMENT COMPLETED SUCCESSFULLY

The WRT300N wireless router was configured with SSID, WPA2-PSK security and DHCP. The laptops obtained IP addresses automatically, connected wirelessly and could ping the router and each other with 0% packet loss.

VIVA QUESTIONS AND ANSWERS

A. Wireless Basics

Q1. What is a wireless network?

Ans: A wireless network connects devices using radio waves instead of cables. Devices such as laptops and phones talk to an access point or wireless router over the air, which gives mobility and removes the need for cabling to every device.

Q2. What is WLAN?

Ans: WLAN (Wireless Local Area Network) is a LAN in which devices communicate wirelessly, usually through an access point. In this experiment, the WRT300N router and the laptops together form a WLAN.

Q3. What is Wi-Fi?

Ans: Wi-Fi is the commercial name for wireless LAN technology based on the IEEE 802.11 standards. It is certified by the Wi-Fi Alliance so that devices of different companies work together.

Q4. What is the IEEE standard for WLAN?

Ans: IEEE 802.11. Its versions include 802.11b (11 Mbps, 2.4 GHz), 802.11a (54 Mbps, 5 GHz), 802.11g (54 Mbps, 2.4 GHz), 802.11n (up to 600 Mbps, 2.4/5 GHz) and 802.11ac/ax for higher speeds.

Q5. What is an access point (AP)?

Ans: An access point is a device that lets wireless clients connect to a wired network. It converts between wireless and wired frames. A wireless router such as the WRT300N has an access point built into it along with routing and DHCP functions.

Q6. What is a wireless router and how is it different from an access point?

Ans: A wireless router combines a router, a switch, an access point and usually a DHCP server in one box, so it can create a network and connect it to the internet. A plain access point only extends an existing wired network to wireless clients and does no routing.

Q7. What is the WRT300N?

Ans: It is a Linksys wireless-N broadband router model that Packet Tracer provides. It supports 802.11n (and b/g) in the 2.4 GHz band, with a built-in switch, DHCP server and security options.

Q8. What is the WPC300N?

Ans: It is the Linksys wireless-N notebook adapter (PC card) that Packet Tracer uses to give a laptop wireless capability. A laptop has a wired Ethernet module by default and must have this module installed to connect to the wireless network.

Q9. Why did we remove the LAN Ethernet module from the laptop?

Ans: The slot has room for only one module, and the default module is wired. To use wireless, the wired module is removed and the wireless WPC300N module is placed in the empty slot, after switching the laptop off.

Q10. Why must the laptop be switched off before changing the module?

Ans: Packet Tracer does not allow adding or removing a module while the device is powered on, just as a real laptop card should not be changed while powered. After inserting the module, it is switched on again so the device detects it.

Q11. What is a wireless infrastructure mode?

Ans: In infrastructure mode, all wireless clients communicate through an access point or router. The other mode is ad hoc, where clients talk directly to each other. The Site Information in our screenshot shows “Infrastructure” as the wireless mode.

Q12. What is ad hoc mode?

Ans: In ad hoc (peer-to-peer) mode, wireless devices connect directly to each other without an access point. It is used for small, temporary networks, and has limited range and security.

Q13. What is the difference between wired and wireless networks?

Ans: Wired networks use cables; they offer higher speed, lower latency, better security and stability, but lack mobility. Wireless networks offer mobility, easy installation and flexibility, but have lower speed, interference, limited range and weaker security if not configured properly.

Q14. What are the advantages and disadvantages of wireless networks?

Ans: Advantages: mobility, no cabling cost, easy to add devices and quick to install. Disadvantages: interference from other devices, limited range, lower throughput, and security risk as the signal can be intercepted.

Q15. Which frequency bands are used by Wi-Fi?

Ans: Mainly 2.4 GHz and 5 GHz (and 6 GHz in Wi-Fi 6E). 2.4 GHz has a longer range but is crowded and slower, while 5 GHz is faster but has a shorter range. This experiment uses the 2.4 GHz band, as shown by the 2.4 GHz logo on the adapter.

Q16. What is a channel? Which channel did we use?

Ans: A channel is a small slice of the frequency band used by the wireless network. The router in our screenshot used channel 6 (2.437 GHz). In the 2.4 GHz band, channels 1, 6 and 11 do not overlap, so they are the commonly used ones.

Q17. Why are channels 1, 6 and 11 preferred in 2.4 GHz?

Ans: Each Wi-Fi channel is about 22 MHz wide while channels are only 5 MHz apart, so most of them overlap. Channels 1, 6 and 11 are spaced far enough apart not to overlap, which reduces interference between neighbouring networks.

Q18. What is the coverage range in the router settings?

Ans: It is the distance in metres within which the router's signal can be received. In Packet Tracer the default is 250 metres. A laptop placed outside this range will not see the SSID or will have a weak signal.

Q19. What is a signal strength percentage?

Ans: It shows how strongly the laptop receives the router's signal. 100% in our screenshot means a very good signal. As the laptop is moved away or obstacles are added, the percentage drops and speed/stability fall.

B. SSID and Security

Q20. What is an SSID?

Ans: SSID (Service Set Identifier) is the name of a wireless network, up to 32 characters, broadcast by the router so that users can identify and choose it. In this experiment it is "Home Router".

Q21. Why do we need to give an SSID?

Ans: Several wireless networks can exist in the same area, so the SSID helps clients pick the right one. A custom SSID also makes the network identifiable and avoids confusion with the default one.

Q22. What is SSID broadcast? Why would we disable it?

Ans: The router regularly announces its SSID so that clients can see it in the list. Disabling it hides the network name, which gives slight obscurity, but it is not real security because the name can still be found with tools. Users must then type the SSID manually.

Q23. Why is security needed in a wireless network?

Ans: Radio signals travel through the air and can be received by anyone nearby. Without security, anyone could connect, use the bandwidth or capture the data. Security provides authentication (who can join) and encryption (data is unreadable to outsiders).

Q24. What are the wireless security types available?

Ans: Disabled (open), WEP, WPA and WPA2, each of which has a Personal (PSK) and an Enterprise form, and the newer WPA3. The WRT300N in Packet Tracer offers Disabled, WEP, WPA-PSK, WPA2-PSK, WPA and WPA2.

Q25. What is WEP? Why is it not recommended?

Ans: Wired Equivalent Privacy is an old security standard that uses the RC4 cipher with a weak 24-bit initialisation vector. It can be cracked within minutes, so it is considered insecure and replaced by WPA/WPA2.

Q26. What is WPA and WPA2?

Ans: Wi-Fi Protected Access (WPA) was an interim fix for WEP using TKIP. WPA2 is the full 802.11i standard and uses the AES-CCMP encryption, which is much stronger. WPA2 is the standard used in this practical.

Q27. What is the difference between WPA2-PSK and WPA2 (Enterprise)?

Ans: WPA2-PSK (Personal) uses one shared password for everybody and is used in homes and small offices. WPA2 Enterprise uses a RADIUS server for authentication, so each user has own credentials; it is more secure and used in large organisations. That is why the RADIUS settings are greyed out in our screenshot.

Q28. What is PSK?

Ans: PSK (Pre-Shared Key) is the password that is configured on the router and entered by the client to join the network. Both sides use it to derive the encryption keys. Here the PSK pass phrase is "Password".

Q29. What is AES?

Ans: Advanced Encryption Standard is a strong symmetric encryption algorithm used with WPA2. The "Encryption Type: AES" in the router settings means the data over the air is encrypted with AES.

Q30. What is TKIP and how is it different from AES?

Ans: Temporal Key Integrity Protocol was used with WPA as a patch for WEP hardware; it still has weaknesses. AES-CCMP is stronger and is required for WPA2, so AES is chosen.

Q31. What happens if the wrong pre-shared key is entered?

Ans: The authentication fails and the laptop does not connect to the network. It does not receive an IP address and cannot ping the router. The correct key must be entered in the Connect dialog.

Q32. What happens if the security is disabled on the router?

Ans: The network becomes open: any laptop can connect without any password, and the traffic is not encrypted. It is acceptable only for testing, not for real use.

Q33. What if the laptop's security setting differs from the router's?

Ans: The two sides cannot negotiate the same encryption method, so the connection fails. The Site Information in the laptop's Connect tab shows the security type (WPA2-PSK) of the router, which the client must match.

Q34. What is MAC address filtering?

Ans: It is a security feature that allows only listed MAC addresses to connect. It is easy to bypass by spoofing, so it is only an additional layer. It was not used in this experiment.

Q35. What is authentication and what is encryption?

Ans: Authentication verifies the identity of the client (e.g. password check) before it is allowed to join. Encryption scrambles the data so that, even if intercepted, it cannot be read. WPA2-PSK provides both.

C. IP Addressing and DHCP

Q36. What is DHCP?

Ans: Dynamic Host Configuration Protocol automatically gives clients an IP address, subnet mask, default gateway and DNS server. It removes the need to configure every device by hand and avoids IP conflicts.

Q37. Why did we use DHCP in this experiment?

Ans: The wireless clients are not fixed and can be many, so the router's DHCP server gives each laptop an IP automatically as soon as it connects. This is simpler than assigning static IPs to every laptop.

Q38. What is the DHCP process (DORA)?

Ans: Discover (client broadcasts a request for a server), Offer (server proposes an IP), Request (client accepts the offer) and Acknowledge (server confirms the lease). After DORA, the laptop has an IP address.

Q39. What does "Internet Connection Type – Automatic Configuration (DHCP)" mean?

Ans: It means the router's Internet (WAN) port gets its own IP address automatically from the ISP's DHCP server. It is the default and is the right choice when there is no static IP from the provider.

Q40. What is the router's IP address 192.168.0.1 used for?

Ans: It is the LAN-side address of the router and acts as the default gateway for all the wireless clients. It is also used to open the router's GUI for configuration.

Q41. Can the router's default IP be changed? Why?

Ans: Yes. It can be set to any private address, such as 192.168.1.1 or 192.168.30.1, to avoid conflicts with another network or to match the addressing plan. The DHCP range must then be changed to the same network.

Q42. What are private IP address ranges?

Ans: 10.0.0.0–10.255.255.255 (Class A), 172.16.0.0–172.31.255.255 (Class B) and 192.168.0.0–192.168.255.255 (Class C). They are not routable on the internet; our 192.168.0.0/24 network is a private Class C network.

Q43. What is the start IP address in DHCP?

Ans: It is the first address of the pool that the router can give to clients. With a start IP of 192.168.0.2 (192.168.0.1 is the router itself), the first laptop gets .2 or the next free address.

Q44. What is the "Maximum number of Users"?

Ans: It is the size of the DHCP pool, i.e. the number of clients which can get an IP address. With start IP .2 and 50 users, the range is 192.168.0.2 – 192.168.0.51. When the pool is exhausted, new devices do not get an IP.

Q45. Why must the DHCP start IP be in the same range as the gateway?

Ans: The clients need to be in the same network as the router to use it as the gateway. If the pool was in a different network, the clients would be unable to reach the gateway or each other.

Q46. What is the client lease time?

Ans: It is the time for which a client may keep the IP address given by DHCP, after which it must renew it. In the screenshot, 0 means one day.

Q47. What is the subnet mask 255.255.255.0 and how many hosts does it allow?

Ans: It is a /24 mask, with 24 network bits and 8 host bits. That allows 2^8 − 2 = 254 usable addresses (from .1 to .254), as .0 is the network and .255 is the broadcast address.

Q48. What is the role of DNS and WINS in the router's settings?

Ans: DNS translates names such as google.com to IP addresses, and WINS translates NetBIOS names (an old Microsoft system). In this practical both are 0.0.0.0 as we do not use name resolution; we ping by IP address.

Q49. What if two laptops get the same IP?

Ans: This is called an IP conflict, and it is normally avoided because DHCP gives each client a unique address from the pool. If one was set manually inside the pool, a conflict could occur and both would have unstable connectivity.

Q50. How can we see the IP address assigned to a laptop?

Ans: Open the Command Prompt on the laptop and type ipconfig (ipconfig /all for more details), or use Desktop → IP Configuration. The IP, mask, gateway and DNS given by DHCP are displayed.

Q51. What if DHCP is disabled on the router?

Ans: Clients still connect to the wireless network, but do not receive an IP address, so they cannot communicate. IPs must be configured manually (static) on each laptop, in the router's network.

D. Packet Tracer Procedure and Verification

Q52. Why did we use the Config tab for security and the GUI tab for DHCP?

Ans: The Config → Wireless page handles the wireless settings (SSID, channel, security). The GUI tab shows the router's own web setup page, where the LAN IP and DHCP server are set, the same as on a real Linksys router.

Q53. Why did we use Save Settings in the GUI?

Ans: The changes in the router's GUI page are applied only after clicking Save Settings at the bottom. If we do not save them, the router keeps the old IP and DHCP values.

Q54. What does the Connect tab of PC Wireless show?

Ans: It lists the wireless networks in range along with the channel, the signal strength and the site information (mode, network type, radio band, security, MAC). We select the network and click Connect.

Q55. What does Link Information show?

Ans: It shows the current state of the connection: "You have successfully connected to the access point", along with the signal strength and link quality bars. It is also used to check that the adapter is active.

Q56. What does "Adapter is Inactive" and "Adapter is Active" mean?

Ans: Inactive (Laptop0 before connecting) means the adapter is not associated with any network. Active (Laptop1 after connecting) means the adapter is connected to the access point and working.

Q57. How do we know the laptop is connected to the router?

Ans: There are three signs: the Link Information shows a successful connection, a dotted wireless link appears between the laptop and the router on the workspace, and the ping to the gateway gives replies.

Q58. What does the dotted line between the laptop and the router represent?

Ans: It is the wireless link. Unlike a cable, there is no physical connection; Packet Tracer draws the dotted line to show that the laptop is associated with the router.

Q59. Why did we ping the router's gateway IP first?

Ans: To check the basic connectivity between the laptop and the router. If this fails, there is no point testing other laptops, because the problem is in the wireless connection, security key or IP settings.

Q60. Why was TTL=255 for the router and TTL=128 for the laptop?

Ans: TTL is the starting value set by the operating system of the replying device. Cisco devices start with 255, and Windows-like PCs/laptops start with 128. Both are on the same network (no router hop), so there is no decrease.

Q61. What is the use of the ping command and ICMP?

Ans: Ping sends ICMP echo requests and waits for echo replies to test reachability and measure round-trip time. In our output, 4 packets were sent, 4 received and 0% lost, so the connection is good.

Q62. How do you read the ping statistics?

Ans: "Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)" shows the reliability. The minimum, maximum and average times are the round-trip delays in milliseconds. Here the average was 13 ms to the router and 18 ms to Laptop1; wireless is a bit slower than wired links.

Q63. Why is wireless ping time higher than wired?

Ans: Wireless needs extra steps such as channel access (CSMA/CA), acknowledgements and encryption, and the radio medium is shared. This increases the delay and makes it variable compared to a wired link.

Q64. What does "Invalid Command" mean in the command prompt?

Ans: The PC command prompt did not recognise the typed command. "show ip" is a Cisco IOS router/switch command, not a PC command. On a laptop, use ipconfig.

Q65. Why can the laptops ping each other?

Ans: Both are connected to the same router (same SSID, same subnet), so the router forwards the traffic between them. They received IPs from the same DHCP pool, so no routing between networks is needed.

Q66. How do we add more laptops?

Ans: Place another Laptop-PT, change its module to WPC300N (power off, swap, power on), open PC Wireless → Connect, select the SSID and enter the key. The router's DHCP then gives it the next IP address.

E. Scenario / "What if" Questions

Q67. What if the SSID is not visible in the Connect tab?

Ans: Possible causes: the laptop is outside the coverage range, the adapter module is not installed or the laptop is off, the SSID broadcast is disabled, or the router is off. Click Refresh and check these.

Q68. What if the laptop is placed outside the coverage range?

Ans: The signal strength becomes weak or the network disappears from the list, and the laptop disconnects. Moving it closer to the router restores the link.

Q69. What if the laptop connects but ping to the gateway fails?

Ans: The wireless connection might be fine but the IP settings are wrong: for example the laptop did not get an IP from DHCP, has a wrong subnet, or the DHCP pool is exhausted. Check with ipconfig.

Q70. What if the laptop shows IP 169.254.x.x (APIPA)?

Ans: It means the laptop could not get an address from a DHCP server and gave itself a link-local address. Check that DHCP is enabled on the router and that the laptop is really connected.

Q71. What if two routers have the same SSID?

Ans: Clients may connect to either one, causing confusion, and in the worst case they connect to the wrong router. Different networks in the same area should have different SSIDs and non-overlapping channels.

Q72. What if two nearby routers use the same channel?

Ans: Their signals interfere with each other (co-channel interference), which reduces speed and increases errors. Using different non-overlapping channels, such as 1, 6 and 11, solves this.

Q73. What if we change the router's IP to 192.168.30.1?

Ans: The DHCP start address and the clients' addresses must also be in 192.168.30.x, and the gateway for the clients becomes 192.168.30.1. The ping must then be done to 192.168.30.1 and to the 192.168.30.x DHCP addresses of the laptops.

Q74. What if the PSK is changed on the router after the laptops connect?

Ans: Laptops would be disconnected when they need to re-associate, and the old key will no longer work. Each laptop must reconnect with the new pass phrase.

Q75. What if the laptop still has the wired module installed?

Ans: It has no wireless adapter, so PC Wireless is not available and it cannot see the SSID. It would need a cable to a router/switch port to connect.

Q76. What if the maximum users is set to 2 and three laptops connect?

Ans: The DHCP pool only has two addresses, so the first two laptops get IPs while the third one connects wirelessly but does not receive an IP (and cannot communicate) until the pool is increased.

Q77. What if the laptop is moved while pinging?

Ans: The signal strength changes; if the laptop moves out of range, replies stop (Request timed out), and when it returns, the connection is restored (it may reconnect automatically).

Q78. What is roaming?

Ans: Roaming is when a client moves from one access point to another with the same SSID without losing the connection. It is used in large buildings and campuses with many access points.

Q79. What is the difference between a hotspot and a router?

Ans: A hotspot is a wireless access point that provides internet to others, often a phone sharing its mobile data; a router is a dedicated network device that routes between networks. Both can offer Wi-Fi to clients.

Q80. What is CSMA/CA?

Ans: Carrier Sense Multiple Access with Collision Avoidance is the method used by Wi-Fi: a device listens before sending, waits a random time and expects an acknowledgement. Wireless devices cannot detect collisions while transmitting, so they avoid them (unlike wired Ethernet, which uses CSMA/CD).

Q81. What is the hidden node problem?

Ans: Two wireless clients can both reach the access point but not each other, so they cannot hear each other's transmissions and may send at the same time, causing collisions at the access point. RTS/CTS is used to reduce this.

Q82. How can the security of a home wireless network be improved?

Ans: Use WPA2 (or WPA3) with a strong passphrase, change the router's default admin password and default IP/SSID, disable WPS, keep firmware updated, turn on MAC filtering if required and switch off the network when not in use.